Finding Unit Rates
Answer Key

Grade 6–7 Ratios and Proportional Relationships

Quick Answers

1. 3	4. 52	7. 0	10. 5
2. 15	5. 3.5	8. 23	
3. 13	6. 28	9. 30	

Curriculum

Level: Grade 6–7 Ratios and Proportional Relationships
6.RP.A.2, 6.RP.A.3 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–4 of 5 across 10 tagged checks.

Common wrong answers

5. *If they got 0.29: divided the pounds by the dollars instead of the dollars by the pounds*
10. *If they got 18: picked the pack with the smaller total price without dividing*

How to use this key. Every solution names the two quantities being compared before it divides, because choosing which quantity goes on the bottom is the whole decision. "Per" always means "divided by the thing that follows", so the quantity named after *per* is the divisor.

1. 8 granola bars cost 24 dollars. Find the cost per bar.
Solution: "Cost per bar" compares dollars to bars, and *per bar* puts bars on the bottom. Scaling the ratio table from 8 bars down to 1 bar means dividing both rows by 8: $24 \div 8 = 3$. Each bar costs $\boxed{\text{3 dollars per bar}}$.

2. 90 pages in 6 minutes. Find pages per minute.
Solution: "Pages per minute" puts minutes on the bottom, so divide the pages by the minutes: $90 \div 6 = 15$. In the table, going from 6 minutes to 1 minute divides the minutes row by 6, so the pages row is divided by 6 too. The printer prints $\boxed{\text{15 pages per minute}}$.

3. Ticket table: 4 tickets cost 52 dollars. Complete the table and give the unit rate.
Solution: From 4 tickets to 1 ticket, divide by 4: $52 \div 4 = 13$ dollars for one ticket. From 4 tickets to 8 tickets, multiply by 2: $52 \times 2 = 104$ dollars, which the unit rate also predicts ($13 \times 8 = 104$). Both directions agree because every column of a ratio table holds the same rate. The unit rate is $\boxed{\text{13 dollars per ticket}}$.

4. 156 miles in 3 hours. Complete the table and give the speed.
Solution: Speed is miles per hour, so hours go on the bottom: $156 \div 3 = 52$ miles in one hour. The 2-hour column is then $52 \times 2 = 104$ miles. The bus travels $\boxed{\text{52 miles per hour}}$.

5. Store A: 5 pounds for 20 dollars. Store B: 8 pounds for 28 dollars. Which is cheaper, and at what rate?
Solution: The totals cannot be compared directly — they buy different amounts — so bring both down to one pound. Store A: $20 \div 5 = 4$ dollars for one pound. Store B: $28 \div 8 = 3.5$ dollars for one pound. Store B is cheaper, by 50 cents on every pound, and its rate is $\boxed{\text{3.50 dollars per pound}}$.

6. 12 muffins from 3 cups of flour. How many muffins from 7 cups?
Solution: First the unit rate: $12 \div 3 = 4$ muffins for one cup. Now scale up to 7 cups: $4 \times 7 = 28$. Written as a proportion, the same work is $3x = 12 \cdot 7$, so $x = 84 \div 3 = 28$. The recipe makes $\boxed{\text{28 muffins}}$.

7. Maya: 12 laps in 4 minutes. Dev: 21 laps in 7 minutes. Subtract Dev's rate from Maya's.
Solution: Maya's rate is $12 \div 4 = 3$ laps per minute. Dev's rate is $21 \div 7 = 3$ laps per minute. Subtracting, $3 - 3$, the difference is $\boxed{0}$. A difference of zero means neither runner is faster — they run at exactly the same rate, even though Dev ran more laps. Comparing lap totals alone would have said Dev was faster by 9 laps, which is the trap this problem is built on.

8. Six boxes hold 138 crayons. Complete the table and give the unit rate.
Solution: Divide both rows by 6 to reach one box: $138 \div 6 = 23$ crayons in one box. The 4-box column follows from the unit rate: $23 \times 4 = 92$ crayons. Each box holds $\boxed{\text{23 crayons per box}}$.

9. A pump moves 84 gallons in 12 minutes. How long for 210 gallons?
Solution: The useful unit rate here is gallons per minute: $84 \div 12 = 7$ gallons each minute. To move 210 gallons the pump needs $210 \div 7 = 30$ minutes. As a proportion this is $84x = 12 \cdot 210$, giving $x = 2520 \div 84 = 30$. The pump needs $\boxed{\text{30 minutes}}$.

10. 3-pack for 18 dollars, 5-pack for 25 dollars. Jae compares 18 and 25. Explain the flaw and give the better buy's price per bottle.
Solution: Jae compared totals for *different numbers of bottles*, which says only that the small pack costs less to buy, not that it is better value. Bring both to one bottle: $18 \div 3 = 6$ dollars per bottle for the small pack, and $25 \div 5 = 5$ dollars per bottle for the large pack. The large pack is the better buy at $\boxed{\text{5 dollars per bottle}}$.